

NGUYEN THANH PHAT

Senior Full-Stack Developer

PHP · Node.js · TypeScript · React

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Ho Chi Minh City, Vietnam

TECHNICAL SKILLS

Languages

PHP, TypeScript, JavaScript

Backend

PHP/Phalcon, Node.js, NestJS, REST APIs

Frontend

React, Next.js, Tailwind CSS

Databases

PostgreSQL, MongoDB, MariaDB/MySQL, Redis

Messaging

RabbitMQ, Apache Kafka, BullMQ

Identity & Security

Keycloak, SSO, JWT, OAuth 2.0/OIDC, RBAC, SpiceDB/ReBAC

DevOps

Linux, Docker, Docker Compose, Nginx, Apache, Git, GitLab CI/CD, PM2

Testing

Vitest, Jest, Playwright, manual testing

SUMMARY

Senior Full-Stack Developer with **10+ years of experience** owning production systems end to end, from requirements and architecture to deployment and support. Deep experience in PHP/Phalcon and modern Node.js/TypeScript development, building backend services with NestJS and web applications with Next.js/React. Built and operated a 24/7 platform managing **250,000+ devices**, with ~200,000 concurrently online and **~5,000 events/minute**.

EXPERIENCE

HONEYNET Joint Stock Company

03/2016 – Present

Tech Lead & Full-Stack Developer

2022 – Present

- Led the full rewrite of a PHP/Phalcon file-storage and identity platform to **NestJS + Next.js**, including architecture and database migration; the new platform serves **5,000+ active users**.
- Designed and built a 24/7 TR-069 platform managing **250,000+ devices**, with ~200,000 concurrently online and **~5,000 events/minute** processed by 20+ RabbitMQ workers.
- Re-architected background processing from a process-per-request model to **RabbitMQ + 20+ workers**, eliminating file-descriptor exhaustion, database connection-pool exhaustion, and CPU/memory overload under production traffic.
- Optimized device queries by limiting retrieved fields and adding targeted indexes, reducing response times from **30+ seconds to a few seconds**.
- Lead three developers while remaining hands-on across architecture, implementation, code review, deployment, troubleshooting and production support.

Full-Stack Developer

03/2016 – 2021

- Designed and built a Wi-Fi controller from the ground up with PHP/Phalcon and MariaDB for device management, RADIUS, monitoring, reporting and integration APIs.
- Built an internal operations platform covering COVID-19 declarations, QR codes, executive scheduling, task management, Face ID integration and attendance reporting; later upgraded core services from PHP 7 to PHP 8.

DOMAINS

Network device management
 Identity & access management
 File storage & sharing
 IoT & device integration
 Face recognition & attendance

EDUCATION

Engineer's Degree
 Computer Networks & Communications
 University of Information Technology – VNU-HCM
 Graduated 2020

LANGUAGES

Vietnamese: Native
 English: Technical reading & written communication

SELECTED PROJECTS

File Storage, Sharing & Identity Platform07/2026 – Present
Tech Lead / Full-Stack Developer · NestJS, Next.js, PostgreSQL, SpiceDB, Keycloak, MinIO, Kafka

- Implemented PostgreSQL Itree folder hierarchies, trash, public links and owner/editor/viewer permissions backed by SpiceDB.
- Built a Keycloak administration backend with bulk Excel user import/update, review workflows, audit logging and email tracking/resend.
- Separated HTTP, worker and cron workloads; used Kafka for cross-application events and built an iframe-based File Picker SDK.

ACS Admin – TR-069 Device Management04/2024 – 09/2026
Tech Lead / Full-Stack Developer · PHP 8.3, Phalcon, MongoDB, GenieACS, RabbitMQ

- Built bulk configuration, diagnostics, network maps, device templates/tags and condition-based alerts via Telegram and email.
- Implemented message identifiers to prevent duplicate processing and used database transactions to maintain consistency across multi-step updates.
- Provided integration APIs used by other departments and systems to consume device data for operational reporting.

Digital Suite for a Ward-Level Government Agency07 – 09/2026
Tech Lead / Full-Stack Developer · NestJS, TypeORM, Socket.IO, Next.js, Keycloak

- Built an IAM service with bidirectional Keycloak synchronization and a Directory API for other applications.
- Built a paperless meeting system covering schedules, documents, attendees, live voting and controlled impersonation.
- Implemented automated testing across three backend systems using Vitest and Playwright.

Face ID – Access Control & Attendance07/2022 – 07/2026
Full-Stack Developer · PHP/Phalcon, MongoDB, RabbitMQ, Redis, Firebase

- Built a platform for face-recognition devices, door locks, snapshots and attendance reporting.
- Developed a multi-tenant version with event triggers, alert conditions, recognition schedules and worker-based bulk processing.

ADDITIONAL EXPERIENCE

Freelance Developer – Part-time2019 – 2022

- Maintained and debugged a primary school management platform deployed across **10+ provinces**.
- Developed PBX integrations and business websites for multiple clients.